Photographed-curve restart: results and limits

Fresh frozen inputs; no historical results pooled. Snapshot **2026-10-17T18:25:20.093Z**. The full selected rational 2-component has 46 curves and 90 directed maps, across five levels (1/3/6/12/24).

Exact field and source model	p=72057980207847001; a=18456521102804593; b=19958176276690439
Curve order / point order	72057979853368804 / 18014494963342201, on every transported fixture
Workload and randomness	10 generators × 10 uniform paired nonzero k values; 100 problems/curve. Linux getrandom, unbiased rejection. Answers are excluded from solver input.
Verified native solves in this snapshot	Rho 4600/4600; BSGS 4600/4600. The complete original native corpus is verified.
IC diagnostics / complete recovery	14720 actual GB systems, including 9200 setup probes, 4600 targets and 920 planted controls; all replayed in Singular. Random setup yield 0/9200; target yield 0/4600; full recoveries 0.

What the measured spread means

All 4600 rho cases are verified. Across individual cases the collision counts vary widely, while the ratio of largest to smallest 100-case curve mean is 1.246519. This descriptive ratio is not a hardness or speedup claim.

All 4600 target GB probes used exactly **217410 counted field multiplications**, while elapsed time ranged from 11.5197 to 266.014 ms. All 45000 instrumented map evaluations used 174 multiplications for each P/Q pair. Timing differences with identical arithmetic counts do not establish a hardness difference. This shared four-CPU host ran two rho and two 4-GiB BSGS workers concurrently; no isolation or fastest-reference claim is earned.

For every observed paired key, BSGS online operation counts agree across generators and curves: **verified**. The search identity is floor(k/m) online giant steps, with m=2^27; table construction is separate. Rho's coordinate-based partition changes after transport and its collision time is random, so its operation spread does not by itself rank intrinsic group hardness. Every map has degree 2, coprime to the prime subgroup order; every nonzero transported P and Q retains that exact order.

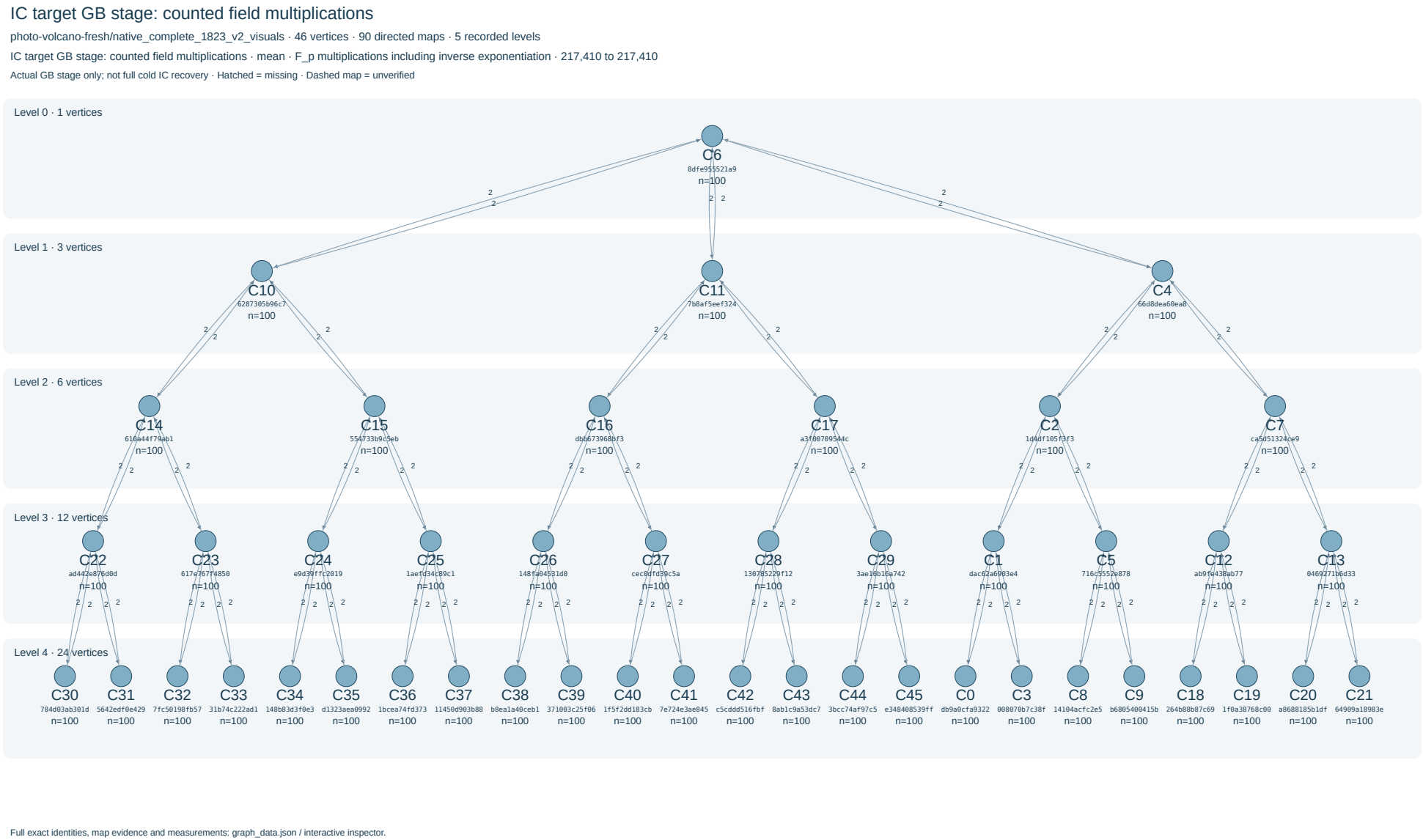
Scope and remaining work

IC uses 12 unknown-log factor-base sign orbits per generator, actual Semaev f3 plus two degree-12 factor polynomials, and 20 random setup trials per generator/curve. Its target Hilbert cutoff is 13, temporary accumulator maximum 23, processed-pair sugar maximum 26; these definitions differ. The audited strict signed two-sum support has only 288 nonzero points per factor base. Transported probes preserve relations and are not independent yield samples. A bounded zero-yield run is not an impossibility proof. A planted full-rank GB-to-linear-algebra control passes separately and adds no random-corpus recoveries.

All original rho and BSGS cases are complete. A separately frozen native arithmetic replay is running; its counters and timings do not replace the original benchmark. Successful random-corpus IC recovery, full event streams, matched isolated repeated timing and an external fastest-tool comparison remain open. Setup/query/verification/destruction and resources are retained privately; these graphs are descriptive stage views, not cold-cost or speed ratios. No complete private archive export is claimed.

IC target GB stage: counted field multiplications

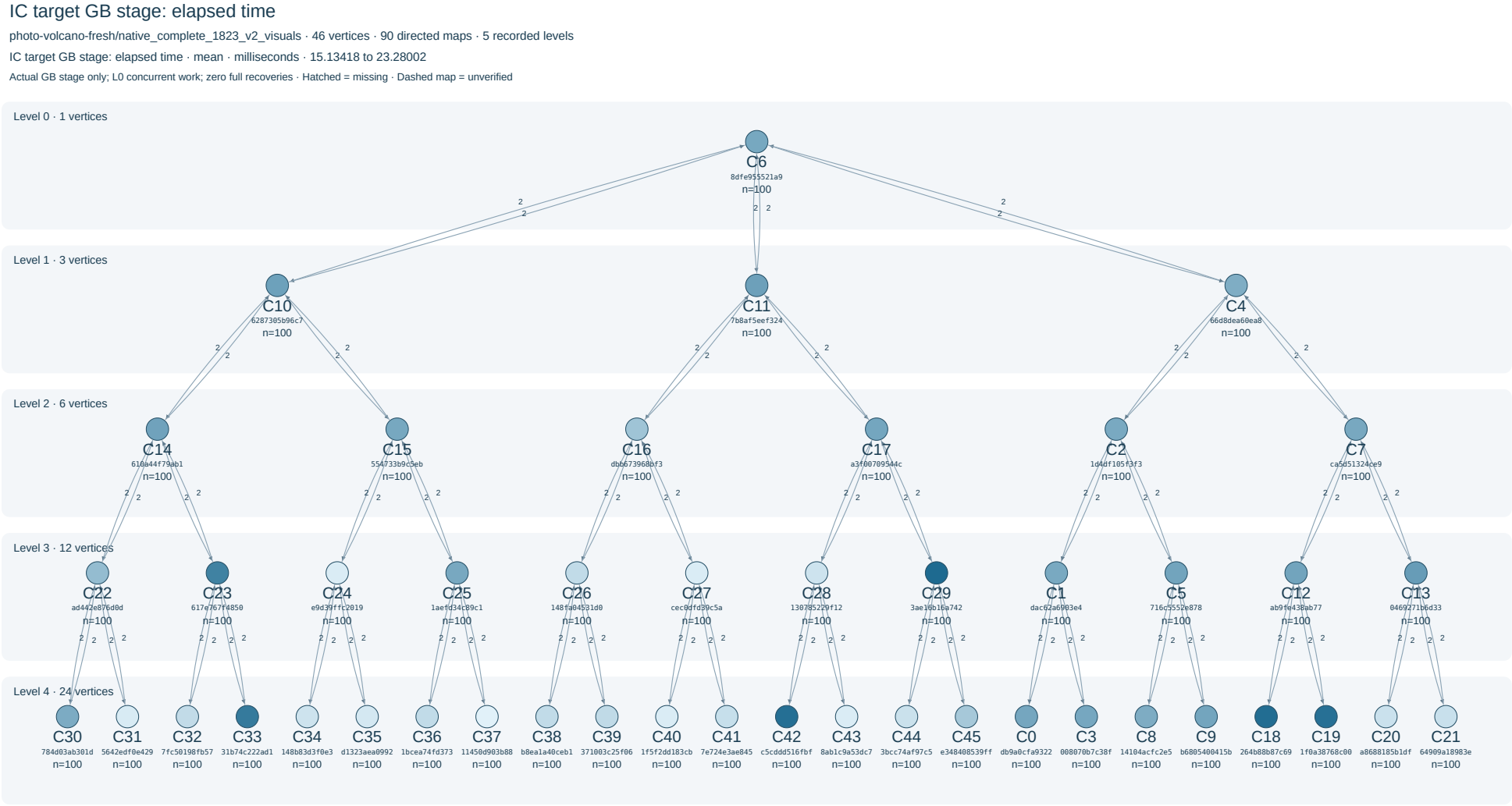
Actual GB stage only; not full cold IC recovery. Unit: F_p multiplications including inverse exponentiation. Node n is the actual measured case count; colors share one range across all five levels. 100 target probes per curve, no full IC recoveries.



Fresh fixture SHA-256: 980d931ce1a804684b160bdab9b213b87688ab9100f0cdf1cd14c390e809acb8. Exact graph and per-node min/mean/max: ic_operations/graph_data.json.

IC target GB stage: elapsed time

Actual GB stage only; L0 concurrent work; zero full recoveries. Unit: milliseconds. Node n is the actual measured case count; colors share one range across all five levels. 100 target probes per curve, no full IC recoveries.



Fresh fixture SHA-256: 980d931ce1a804684b160bdab9b213b87688ab9100f0cdf1cd14c390e809acb8. Exact graph and per-node min/mean/max: ic_wall/graph_data.json.

Fully paired rho problems at this snapshot

Each row is the same transported problem and paired walk seed on all 46 curves. This is an operation-count sample across curves, not 46 independent repeated timing runs.

Problem	Curves	Minimum group operations	Mean group operations	Maximum group operations
I0	46	14,220,700	132,532,000	392,043,000
I1	46	20,173,400	145,277,000	305,616,000
I2	46	9,537,080	95,581,600	293,103,000
I3	46	27,404,300	118,255,000	255,697,000
I4	46	26,293,200	132,504,000	318,197,000
I5	46	8,194,820	114,197,000	295,706,000
I6	46	15,277,000	123,020,000	366,208,000
I7	46	25,746,000	115,634,000	281,960,000
I8	46	19,929,200	124,362,000	268,326,000
I9	46	14,622,700	126,273,000	328,579,000
I10	46	9,155,860	116,352,000	255,173,000
I11	46	20,212,600	111,047,000	258,611,000
I12	46	3,759,520	111,359,000	251,097,000
I13	46	21,471,400	108,773,000	277,235,000
I14	46	35,327,600	133,859,000	245,430,000
I15	46	17,629,200	113,461,000	267,640,000
I16	46	35,386,000	121,814,000	272,133,000
I17	46	13,790,100	134,961,000	360,576,000
I18	46	22,052,200	124,437,000	256,319,000
I19	46	28,634,500	132,387,000	317,200,000
I20	46	7,411,300	122,575,000	292,039,000
I21	46	15,416,000	112,869,000	247,663,000
I22	46	32,403,000	115,619,000	269,655,000
I23	46	37,497,900	118,800,000	267,329,000

Problem	Curves	Minimum group operations	Mean group operations	Maximum group operations
I24	46	22,005,000	124,004,000	262,019,000
I25	46	7,537,500	133,119,000	317,227,000
I26	46	13,057,900	106,024,000	301,189,000
I27	46	15,275,200	128,967,000	290,296,000
I28	46	27,981,600	130,035,000	414,192,000
I29	46	23,802,900	119,137,000	206,661,000
I30	46	6,235,080	124,078,000	282,062,000
I31	46	18,350,400	115,587,000	227,096,000
I32	46	8,544,350	110,351,000	347,777,000
I33	46	33,057,300	122,896,000	218,129,000
I34	46	26,847,800	130,997,000	249,903,000
I35	46	9,369,970	134,342,000	310,301,000
I36	46	20,518,600	115,228,000	327,883,000
I37	46	35,665,900	114,213,000	229,670,000
I38	46	9,088,860	123,467,000	267,873,000
I39	46	4,510,140	126,165,000	277,148,000
I40	46	14,528,900	127,765,000	275,644,000
I41	46	15,280,200	130,736,000	278,747,000
I42	46	19,524,800	125,411,000	268,145,000
I43	46	35,231,900	122,017,000	274,875,000
I44	46	26,234,100	118,224,000	263,676,000
I45	46	35,429,300	118,950,000	248,165,000
I46	46	12,557,600	111,603,000	238,290,000
I47	46	3,962,980	118,079,000	279,201,000
I48	46	42,198,700	144,406,000	429,008,000
I49	46	26,096,500	130,509,000	325,922,000

Problem	Curves	Minimum group operations	Mean group operations	Maximum group operations
I50	46	8,989,820	116,051,000	259,401,000
I51	46	10,956,800	123,522,000	281,206,000
I52	46	8,288,090	122,282,000	256,498,000
I53	46	9,107,490	112,579,000	245,173,000
I54	46	19,942,600	129,827,000	262,832,000
I55	46	11,656,800	115,584,000	233,532,000
I56	46	12,882,400	116,119,000	300,947,000
I57	46	20,590,600	129,514,000	299,955,000
I58	46	21,980,200	126,299,000	298,022,000
I59	46	15,743,800	108,140,000	297,702,000
I60	46	7,867,210	123,640,000	277,927,000
I61	46	12,632,800	119,886,000	322,164,000
I62	46	8,682,710	122,291,000	254,773,000
I63	46	12,902,100	110,367,000	286,810,000
I64	46	20,161,300	123,535,000	282,337,000
I65	46	23,820,300	125,297,000	268,398,000
I66	46	26,486,600	143,605,000	317,686,000
I67	46	17,502,500	114,856,000	297,225,000
I68	46	8,963,210	117,793,000	288,347,000
I69	46	24,497,000	128,770,000	252,188,000
I70	46	42,991,600	117,967,000	229,695,000
I71	46	32,427,000	129,700,000	270,479,000
I72	46	15,817,300	118,108,000	256,838,000
I73	46	20,752,500	101,242,000	205,010,000
I74	46	41,974,800	128,079,000	256,693,000
I75	46	11,737,600	126,566,000	289,528,000

Problem	Curves	Minimum group operations	Mean group operations	Maximum group operations
I76	46	38,081,600	126,195,000	250,670,000
I77	46	15,467,500	110,443,000	239,356,000
I78	46	36,250,000	130,028,000	305,834,000
I79	46	12,760,500	107,786,000	218,591,000
I80	46	10,948,800	115,194,000	367,668,000
I81	46	25,304,500	117,442,000	331,668,000
I82	46	15,876,400	134,968,000	260,347,000
I83	46	35,890,300	124,503,000	256,506,000
I84	46	20,515,400	112,769,000	259,596,000
I85	46	12,566,100	133,978,000	274,379,000
I86	46	15,996,400	112,180,000	283,822,000
I87	46	31,320,000	129,727,000	266,926,000
I88	46	17,747,000	129,407,000	305,835,000
I89	46	12,529,500	106,684,000	234,867,000
I90	46	21,764,900	125,347,000	283,158,000
I91	46	26,863,000	130,405,000	243,807,000
I92	46	4,246,150	120,101,000	280,628,000
I93	46	31,340,300	118,213,000	218,842,000
I94	46	24,744,900	129,403,000	274,509,000
I95	46	16,086,600	118,055,000	265,031,000
I96	46	7,799,660	113,211,000	310,669,000
I97	46	8,386,400	113,208,000	275,749,000
I98	46	17,205,400	110,348,000	239,191,000
I99	46	25,402,600	115,038,000	329,466,000

Reproduce, inspect and retain

Source: source/quantitative_report.mjs, source/measurement_stats.mjs and the multilevel diagram skill. The source-linked topology report contains exact curve/map validation. Native backend commit: b7352eec52676b7b82a220ba296661e8ba878915. Shortlist commit: ef18058588077ef805c55c5dc25140b6bde2046b. Per-curve and per-problem JSON/CSV retain population, min/mean/max and missing values. IC per-problem summaries cover all 100 targets over all 46 curves; map summaries cover all 90 directed edges and 100 instances.

Raw private sources: runs/rho, runs/bsgs, runs/ic; data/map_measurements.jsonl. All attempts and input randomness remain local. Public replay inputs omit expected/recovered scalars and raw RNG journals. The public polynomial-system archive exports only mathematical systems, returned bases, factor-base points and aggregate diagnostic metrics. Independent audit: validation/singular_audit.json. Positive control: validation/ic_full_pipeline_control.json. Snapshot verification: validation/checkpoint_01_verification.json.

Host: Intel Xeon Platinum 8573C, Linux x86-64, four-CPU cgroup quota, 16 GiB memory, no GPU; L0 concurrent timing. Unmodified native group/BSGS/rho tests pass 376415/2063/134 checks. CMake, Cargo and Conductor are unavailable; GCC direct build succeeds. No whole-Rust-library check pass is claimed.

Canonical curves and progress scoreboards retain their prior claims: this fresh measurement adds no admitted performance or boundary promotion. The earlier frozen report snapshots remain immutable. This run's benchmark mirror contains identical reports and visuals.